Lab 16 – Database Design and Queries: Schema Design and SQL Generation

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TASK 1 – Student Information System Schema

Prompt:

Design a relational database schema for a Student Information System.

Requirements:

- Tables: Students, Courses, Enrollments
- Define primary keys and foreign keys
- Ensure proper relationships (many-to-many between students and courses) **Generate:**

➢ SQL CREATE TABLE statements
➢ SQL query to insert a new student record
➢ SQL query to fetch all courses enrolled by a specific student

➢ SQL query to count number of students in each course
Ensure the schema is normalized (up to 3NF).

Code:

```
-- STUDENT INFORMATION SYSTEM
-- Drop tables if already exist (optional)
DROP TABLE IF EXISTS Enrollments;
```

```
DROP TABLE IF EXISTS Students;
DROP TABLE IF EXISTS Courses;
-- Create Tables
CREATE TABLE Students (      student_id INTEGER
PRIMARY KEY AUTOINCREMENT,      first_name TEXT
NOT NULL,      last_name TEXT NOT NULL,



    email TEXT UNIQUE NOT NULL
);
CREATE TABLE Courses (
    course_id INTEGER PRIMARY KEY AUTOINCREMENT,      course_name
TEXT NOT NULL,

    course_code TEXT UNIQUE NOT NULL
);
CREATE TABLE Enrollments (
    enrollment_id INTEGER PRIMARY KEY AUTOINCREMENT,
student_id INTEGER,
    course_id INTEGER,      FOREIGN KEY (student_id)
REFERENCES Students(student_id),

    FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);
```

```sql
-- Insert Sample Students
INSERT INTO Students (first_name, last_name, email) VALUES
('John', 'Doe', 'john.doe@example.com'),
('Alice', 'Smith', 'alice.smith@example.com'),
('Bob', 'Johnson', 'bob.johnson@example.com');
-- Insert Sample Courses

INSERT INTO Courses (course_name, course_code) VALUES
('Database Management', 'DBMS101'),
('Computer Networks', 'CN102'),

('Operating Systems', 'OS103'),
('Data Structures', 'DS104');
-- Insert Enrollments

INSERT INTO Enrollments (student_id, course_id) VALUES
(1, 1),

(1, 2),
(2, 1),
(2, 3),

(3, 2),
(3, 4);

-- Query 1: Fetch all courses of a student
SELECT s.first_name, s.last_name, c.course_name
FROM Students s
JOIN Enrollments e ON s.student_id = e.student_id
JOIN Courses c ON c.course_id = e.course_id
WHERE s.email = 'john.doe@example.com';
-- Query 2: Count students in each course

SELECT c.course_name, COUNT(e.student_id) AS student_count

FROM Courses c
LEFT JOIN Enrollments e ON c.course_id = e.course_id
GROUP BY c.course_name;
```

Output:

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first_name	last_name	course_name
John	Doe	Database Management
John	Doe	Computer Networks

SQL ▾ 1 / 1 1 - 4 of 4

course_name	student_count
Computer Networks	2
Data Structures	1
Database Management	2
Operating Systems	1

TASK 2 – Hospital Management Database

Prompt:

Design a relational database schema for a Hospital Management System.

Requirements:

- Tables: Doctors, Patients, Appointments
- Include constraints like primary keys, foreign keys, unique IDs, and valid date fields
- Ensure normalization (up to 3NF) **Generate:**
  - ➢ SQL CREATE TABLE statements
  - ➢ Query to list all appointments for a specific doctor

- Query to retrieve patient history by patient ID
- Query to count total patients treated by each doctor Use JOIN operations where necessary.

Code:

```
-- HOSPITAL MANAGEMENT SYSTEM
-- Drop tables if already exist
DROP TABLE IF EXISTS Appointments;

DROP TABLE IF EXISTS Patients;
DROP TABLE IF EXISTS Doctors;
-- Create Tables  CREATE
TABLE Doctors (
    doctor_id INTEGER PRIMARY KEY AUTOINCREMENT,

    doctor_name TEXT NOT NULL,
specialization TEXT NOT NULL,

    email TEXT UNIQUE
);
CREATE TABLE Patients (

    patient_id INTEGER PRIMARY KEY AUTOINCREMENT,
patient_name TEXT NOT NULL,

    age INTEGER CHECK(age > 0),
gender TEXT,
```

```
    phone TEXT UNIQUE
);
CREATE TABLE Appointments (

    appointment_id INTEGER PRIMARY KEY AUTOINCREMENT,
doctor_id INTEGER,      patient_id INTEGER,


    appointment_date TEXT NOT NULL,      diagnosis TEXT,
```

```sql
    FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id),
    FOREIGN KEY (patient_id) REFERENCES Patients(patient_id) );
-- Insert Sample Doctors


INSERT INTO Doctors (doctor_name, specialization, email) VALUES
('Dr. Rajesh Kumar', 'Cardiologist', 'rajesh@hospital.com'),
('Dr. Priya Sharma', 'Neurologist', 'priya@hospital.com'),
('Dr. Amit Verma', 'Orthopedic', 'amit@hospital.com');
-- Insert Sample Patients
INSERT INTO Patients (patient_name, age, gender, phone) VALUES
('John Doe', 25, 'Male', '9876543210'),
('Alice Smith', 30, 'Female', '9876543211'),
('Bob Johnson', 40, 'Male', '9876543212');
-- Insert Appointments
INSERT INTO Appointments (doctor_id, patient_id, appointment_date,
diagnosis) VALUES

(1, 1, '2026-03-01', 'Heart Checkup'),
(1, 2, '2026-03-02', 'Chest Pain'),
(2, 1, '2026-03-03', 'Migraine'),
(3, 3, '2026-03-04', 'Fracture'),
(2, 2, '2026-03-05', 'Headache');
-- Query 1: List all appointments for a specific doctor
-- (Example: Doctor ID = 1)
SELECT d.doctor_name, p.patient_name, a.appointment_date, a.diagnosis
FROM Appointments a
JOIN Doctors d ON a.doctor_id = d.doctor_id
JOIN Patients p ON a.patient_id = p.patient_id
WHERE d.doctor_id = 1;
-- Query 2: Retrieve patient history by patient ID
-- (Example: Patient ID = 1)
SELECT p.patient_name, d.doctor_name, a.appointment_date, a.diagnosis
FROM Appointments a
JOIN Patients p ON a.patient_id = p.patient_id
JOIN Doctors d ON a.doctor_id = d.doctor_id
WHERE p.patient_id = 1;
```

```sql
-- Query 3: Count total patients treated by each doctor
SELECT d.doctor_name, COUNT(DISTINCT a.patient_id) AS total_patients
FROM Doctors d
LEFT JOIN Appointments a ON d.doctor_id = a.doctor_id
GROUP BY d.doctor_name;
```

Output:

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doctor_name	patient_name	appointment_date	diagnosis
Dr. Rajesh Kumar	John Doe	2026-03-01	Heart Checkup
Dr. Rajesh Kumar	Alice Smith	2026-03-02	Chest Pain

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patient_name	doctor_name	appointment_date	diagnosis
John Doe	Dr. Rajesh Kumar	2026-03-01	Heart Checkup
John Doe	Dr. Priya Sharma	2026-03-03	Migraine

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doctor_name	total_patients
Dr. Amit Verma	1
Dr. Priya Sharma	2
Dr. Rajesh Kumar	2

TASK 3 – Library Database

Prompt:

Design a relational database schema for a Library Management System.

Requirements:

- Tables: Books, Members, Loans
- Define primary and foreign keys
- Ensure normalization (up to 3NF)
- Suggest indexing strategies for faster query performance

Generate:

- ➢ SQL CREATE TABLE statements
- ➢ Query to retrieve all books currently issued
- ➢ Query to find overdue books (loan date > 30 days)
- ➢ Query to count number of books loaned by each member Include JOINs and WHERE conditions.

Code:

```
DROP TABLE IF EXISTS Loans;
DROP TABLE IF EXISTS Members;
DROP TABLE IF EXISTS Books;
CREATE TABLE Books (      book_id INTEGER
PRIMARY KEY AUTOINCREMENT,      title TEXT NOT
NULL,      author TEXT NOT NULL,      isbn
TEXT UNIQUE,
    available_copies INTEGER NOT NULL CHECK(available_copies >= 0)


);
CREATE TABLE Members (      member_id INTEGER
PRIMARY KEY AUTOINCREMENT,      member_name TEXT
NOT NULL,
```

```
    email TEXT UNIQUE,
    phone TEXT UNIQUE
```

```
);
CREATE TABLE Loans (
    loan_id INTEGER PRIMARY KEY AUTOINCREMENT,
    book_id INTEGER,     member_id
INTEGER,
```

```
    loan_date TEXT NOT NULL,
return_date TEXT,
    FOREIGN KEY (book_id) REFERENCES Books(book_id),
    FOREIGN KEY (member_id) REFERENCES Members(member_id)  );
CREATE INDEX idx_loans_book_id ON Loans(book_id);
CREATE INDEX idx_loans_member_id ON Loans(member_id); CREATE
INDEX idx_loans_loan_date ON Loans(loan_date);
INSERT INTO Books (title, author, isbn, available_copies) VALUES
('Database System Concepts', 'Silberschatz', 'ISBN001', 5),
('Computer Networks', 'Andrew Tanenbaum', 'ISBN002', 3),
('Operating Systems', 'Galvin', 'ISBN003', 4),


('Data Structures', 'Seymour Lipschutz', 'ISBN004', 2);
INSERT INTO Members (member_name, email, phone) VALUES
('John Doe', 'john@example.com', '9000000001'),
('Alice Smith', 'alice@example.com', '9000000002'),
('Bob Johnson', 'bob@example.com', '9000000003');
INSERT INTO Loans (book_id, member_id, loan_date, return_date) VALUES
(1, 1, '2026-02-20', NULL),
(2, 1, '2026-02-15', '2026-02-25'),
(3, 2, '2026-02-10', NULL),  (4,
3, '2026-01-01', NULL),

(2, 2, '2026-01-15', '2026-02-20');
SELECT b.title, m.member_name, l.loan_date
```

```sql
FROM Loans l
JOIN Books b ON l.book_id = b.book_id
JOIN Members m ON l.member_id = m.member_id
WHERE l.return_date IS NULL;
SELECT b.title, m.member_name, l.loan_date

FROM Loans l
JOIN Books b ON l.book_id = b.book_id
JOIN Members m ON l.member_id = m.member_id
WHERE l.return_date IS NULL
AND DATE(l.loan_date) <= DATE('now', '-30 days');
SELECT m.member_name, COUNT(l.loan_id) AS total_books

FROM Members m
LEFT JOIN Loans l ON m.member_id = l.member_id
GROUP BY m.member_name;
```

Output:

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title	member_name	loan_date
Database System Concepts	John Doe	2026-02-20
Operating Systems	Alice Smith	2026-02-10
Data Structures	Bob Johnson	2026-01-01

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title	member_name	loan_date
Database System Concepts	John Doe	2026-02-20
Operating Systems	Alice Smith	2026-02-10
Data Structures	Bob Johnson	2026-01-01

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member_name	total_books
Alice Smith	2
Bob Johnson	1
John Doe	2

TASK 4 – Real-Time Application: Online Shopping Database

Prompt:

Design a relational database schema for an E-commerce platform.
Requirements:

- Tables: Users, Products, Orders, OrderDetails
- Define primary keys, foreign keys, and relationships
- Ensure normalization (up to 3NF)
- Suggest query optimization techniques **Generate:**

➢ SQL CREATE TABLE statements
➢ Query to retrieve all orders by a user
➢ Query to find the most purchased product
➢ Query to calculate total revenue in a given month

Include aggregation functions, JOINs, and optimization suggestions.

Code:

```
DROP TABLE IF EXISTS OrderDetails;

DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Users;

CREATE TABLE Users (
    user_id INTEGER PRIMARY KEY AUTOINCREMENT,

    user_name TEXT NOT NULL,

    email TEXT UNIQUE NOT NULL,
phone TEXT UNIQUE );  CREATE
TABLE Products (
```

```
    product_id INTEGER PRIMARY KEY AUTOINCREMENT,
product_name TEXT NOT NULL,      price REAL NOT
NULL CHECK(price > 0),

    stock INTEGER NOT NULL CHECK(stock >= 0)
);
CREATE TABLE Orders (
    order_id INTEGER PRIMARY KEY AUTOINCREMENT,
user_id INTEGER,      order_date TEXT NOT NULL,
    FOREIGN KEY (user_id) REFERENCES Users(user_id)


);
CREATE TABLE OrderDetails (      order_detail_id
INTEGER PRIMARY KEY AUTOINCREMENT,      order_id
INTEGER,      product_id INTEGER,


    quantity INTEGER NOT NULL CHECK(quantity > 0),
```

```
    FOREIGN KEY (order_id) REFERENCES Orders(order_id),
    FOREIGN KEY (product_id) REFERENCES Products(product_id)  );
CREATE INDEX idx_orders_user_id ON Orders(user_id);
CREATE INDEX idx_orderdetails_order_id ON OrderDetails(order_id);
CREATE INDEX idx_orderdetails_product_id ON OrderDetails(product_id);
INSERT INTO Users (user_name, email, phone) VALUES


('John Doe', 'john@example.com', '9000000001'),
('Alice Smith', 'alice@example.com', '9000000002'),
('Bob Johnson', 'bob@example.com', '9000000003');
INSERT INTO Products (product_name, price, stock) VALUES
('Laptop', 60000, 10),
('Mobile Phone', 20000, 20),
('Headphones', 2000, 50),
('Keyboard', 1500, 30);
INSERT INTO Orders (user_id, order_date) VALUES
(1, '2026-03-01'),
(1, '2026-03-05'),
(2, '2026-03-10'),
```

```
(3, '2026-02-20');
INSERT INTO OrderDetails (order_id, product_id, quantity) VALUES
(1, 1, 1),
(1, 3, 2),
(2, 2, 1),
(2, 4, 1),
(3, 2, 2),  (4,
3, 3);

SELECT u.user_name, o.order_id, o.order_date
FROM Users u
JOIN Orders o ON u.user_id = o.user_id
WHERE u.user_id = 1;
SELECT p.product_name, SUM(od.quantity) AS total_quantity
FROM OrderDetails od
JOIN Products p ON od.product_id = p.product_id
GROUP BY p.product_name
ORDER BY total_quantity DESC
LIMIT 1;
SELECT SUM(p.price * od.quantity) AS total_revenue
FROM OrderDetails od
JOIN Products p ON od.product_id = p.product_id
JOIN Orders o ON od.order_id = o.order_id
WHERE strftime('%Y-%m', o.order_date) = '2026-03';
```

Output:

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user_name	order_id	order_date
John Doe	1	2026-03-01
John Doe	2	2026-03-05

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product_name	total_quantity
Headphones	5

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total_revenue
125500.0

TASK 5 – University Examination Database

Prompt:

Design a relational database schema for a University Examination System.

Requirements:

- Tables: Students, Subjects, Exams, Results
- Define primary keys, foreign keys, and referential integrity constraints
- Ensure normalization (up to 3NF) **Generate:**
  - ➢ SQL CREATE TABLE statements
  - ➢ Query to insert examination results for a student
  - ➢ Query to retrieve subject-wise marks for a student
  - ➢ Query to calculate average marks for each subject Use JOINs and aggregate functions.

Code:

```
DROP TABLE IF EXISTS Results;

DROP TABLE IF EXISTS Exams;

DROP TABLE IF EXISTS Subjects;

DROP TABLE IF EXISTS Students;  CREATE TABLE
Students (
     student_id INTEGER PRIMARY KEY AUTOINCREMENT,
student_name TEXT NOT NULL,      email TEXT UNIQUE
);
CREATE TABLE Subjects (
    subject_id INTEGER PRIMARY KEY AUTOINCREMENT,



    subject_name TEXT NOT NULL,      subject_code TEXT
UNIQUE NOT NULL );


CREATE TABLE Exams (

    exam_id INTEGER PRIMARY KEY AUTOINCREMENT,
subject_id INTEGER,      exam_date TEXT NOT NULL,


    FOREIGN KEY (subject_id) REFERENCES Subjects(subject_id)  );
CREATE TABLE Results (
    result_id INTEGER PRIMARY KEY AUTOINCREMENT,      student_id
INTEGER,



    exam_id INTEGER,

    marks INTEGER CHECK(marks >= 0 AND marks <= 100),
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
    FOREIGN KEY (exam_id) REFERENCES Exams(exam_id)  );
INSERT INTO Students (student_name, email) VALUES
('John Doe', 'john@example.com'),
```

```sql
('Alice Smith', 'alice@example.com'),
('Bob Johnson', 'bob@example.com');
INSERT INTO Subjects (subject_name, subject_code) VALUES
('Mathematics', 'MATH101'),
('Physics', 'PHY102'),
('Computer Science', 'CS103');
INSERT INTO Exams (subject_id, exam_date) VALUES
(1, '2026-03-01'), (2,
'2026-03-02'),
(3, '2026-03-03');
INSERT INTO Results (student_id, exam_id, marks) VALUES
(1, 1, 85),
(1, 2, 78),
(1, 3, 90),
(2, 1, 88),
(2, 2, 76),
(3, 3, 92);
SELECT * FROM Results;
SELECT s.student_name, sub.subject_name, r.marks
FROM Results r
JOIN Students s ON r.student_id = s.student_id
JOIN Exams e ON r.exam_id = e.exam_id
JOIN Subjects sub ON e.subject_id = sub.subject_id
WHERE s.student_id = 1;
SELECT sub.subject_name, AVG(r.marks) AS average_marks
FROM Results r
JOIN Exams e ON r.exam_id = e.exam_id
JOIN Subjects sub ON e.subject_id = sub.subject_id
GROUP BY sub.subject_name;
```

Output:

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result_id	student_id	exam_id	marks
1	1	1	85
2	1	2	78
3	1	3	90
4	2	1	88
5	2	2	76
6	3	3	92

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student_name	subject_name	marks
John Doe	Mathematics	85
John Doe	Physics	78
John Doe	Computer Science	90

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subject_name	average_marks
Computer Science	91.0
Mathematics	86.5
Physics	77.0